

Optimising training. Using FP16 precision (instead of FP32) speeds up training and reduces memory consumption.

```
training_args = TrainingArguments(fp16=True, output_dir="./results")
```

Checkpointing. Saving checkpoints ensures training can resume after interruptions.

```
training_args = TrainingArguments(save_steps=500, save_total_limit=3, output_dir="./results")
```

Evaluation and deployment of machine learning models

Evaluating a machine learning model is a crucial step to ensure its performance and reliability before deployment. Various metrics and benchmark datasets are used to assess the effectiveness of the model.

Evaluation metrics. Depending on the type of model and task, different evaluation metrics are used.

- **Perplexity.** Commonly used in language models, Perplexity measures how well a probability distribution predicts a sample.
- **BLEU (Bilingual Evaluation Understudy).** Evaluates the quality of machine-generated translations against human references.
- **ROUGE (Recall-Oriented Understudy for Gisting Evaluation).** Measures the overlap of n-grams between the model's output and a reference text, often used for text summarisation.

- **Accuracy, Precision, Recall, F1-score.** Used in classification models to measure correctness and performance.

- **Mean Squared Error (MSE) and R-squared.** Applied in regression tasks to measure prediction errors.

Benchmark datasets. Benchmark datasets provide standardised comparisons for model evaluation (Table 5).

Once a model is evaluated and optimised, it is deployed to serve real-world applications. Several tools and frameworks facilitate deployment (Table 6).

Building your own LLM may seem daunting, but with the right open source tools, the process is more accessible than ever. The key is to start small—experiment with fine-tuning an existing model before scaling up to full-fledged training.

Whether you're building a proprietary AI assistant, a specialised content generator, or a research-driven model, the open source community has provided everything you need.

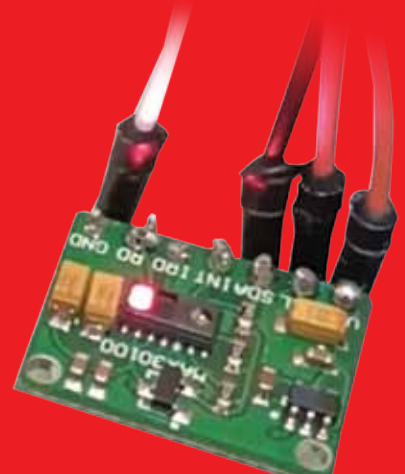
Ready to build your LLM? Choose your framework, set up your environment, and start training today! The future of AI is open source—be part of the revolution! **EFY**

This article was first published in April 2025 issue of Open Source For You magazine.

IOT DIY KIT: Health Tech & Medical Devices

Want to create exciting IoT projects related to medical electronics, medical devices, health or wellness tech?

This DIY kit lets you build smart IoT projects with WiFi connectivity and built-in sensors for monitoring, alerts, and tracking.



POWERED BY



IndusBoard
COIN

Super small sized development board that enables you to create solutions for many use-cases where small size is super important.



SPECIAL INTRO PRICE

₹2,049 (Inclusive of GST)

KNOW MORE / BUY NOW: efy.co.in/healthiotkit

Developed by the Electronics For You team, for you.

DEALERS: • Om Technology Centre, Bengaluru (Mob: +919845400864)
• Bharathi Electronics: Coimbatore (Mob: +919952228618)